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The Editor-in-Chief
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Dear Editor-in-Chief,

We respectfully submit our manuscript, **"NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction,"** for consideration as an original research article in *IEEE Transactions on Multimedia* (TMM).

Scope alignment. NARCIS addresses a core multimedia communications problem: how to transmit authenticated messages by selecting and sending unchanged natural images from a shared index, rather than by modifying a carrier. The technical pipeline is a multimedia processing contribution — a compact self-supervised encoder that maps images to channel-invariant descriptors, a quantile codebook that converts the descriptor space into balanced message symbols, and a multi-attack qualification stage that retains only images whose labels remain stable under channel distortions. These components sit squarely within TMM's scope on multimedia content analysis, multimedia security, and image-based communications. The detectability evaluation (three detector classes, five partitions per dataset, calibration and holdout conditions) also directly addresses the multimedia security analysis that TMM regularly publishes.

Key technical contributions.

- **Feasible-codebook operating point (Contribution 1).** Existing coverless systems report nominal capacity or average symbol accuracy, neither of which verifies that a given message can actually be transmitted from a finite, qualified index. NARCIS introduces the *feasible operating point*: the largest symbol alphabet for which every attack-qualified bucket satisfies the multiplicity required by every protected coded message. Over 1,575 complete-message trials on BOSSBase 1.01 (grayscale) and Caltech-101 (RGB), NARCIS recovered 100% of authenticated plaintexts under 21 channel conditions, including eight unseen holdout strengths.
- **End-to-end coverless pipeline (Contribution 2).** A self-supervised encoder (variance–invariance–covariance objective) produces 64-dimensional normalised descriptors. A quantile partition of the first principal component yields balanced symbol classes. Multi-attack qualification (12 calibration transforms) removes geometrically unstable images; a calibration-locked projection search maintains 210/210 recoveries while raising the minimum stable bucket from 92 to 161 (BOSSBase seed 11). AES-GCM authenticates payloads and session metadata; a keyed Gray permutation limits symbol exposure; Reed–Solomon coding corrects residual label errors. Net throughput scales from 0.184 to 1.213 bits per cover for 8- to 128-byte payloads.
- **Detectability and ablation evaluation (Contribution 3).** Selection bias is quantified with three detector classes (global logistic regression on dense residuals, residual-feature ExtraTrees, selection CNN) over five partitions each of BOSSBase and Caltech-101. BOSSBase AUCs include chance. Caltech-101 classical detectors identify a weak but reproducible selection shift (global logistic AUC 0.531, 95% CI [0.504, 0.557]; ExtraTrees 0.533, [0.504, 0.562]); 20 repeated CNN fits on the most sensitive partition confirm a partition-specific signal (mean AUC 0.533, 95% CI [0.517, 0.548]), reported transparently rather than dismissed. Removing attack qualification on BOSSBase seed 11 reduces complete-message success from 210/210 to 142/210 (67.6%), establishing qualification as the dominant robustness mechanism.

Novelty and positioning. Selection-based coverless systems have been evaluated at the descriptor level — average symbol accuracy, nominal capacity — but have not verified message-level feasibility over a finite qualified index, nor integrated payload authentication, replay control, and error correction into a single testable system. TMM has published related work in learned descriptors for coverless steganography (Ren–Wu 2024; Guo–Ping 2026) and in multimedia security evaluation. NARCIS complements this body of work by introducing the feasible-codebook criterion as the primary evaluation unit and providing systematic cross-dataset, multi-detector detectability evidence.

On capacity, the paper distinguishes two metrics that are not interchangeable: *nominal symbol capacity* ($\log_2 K$ bpc, measured at the symbol level before any protocol overhead) and *net authenticated throughput* (plaintext bpc after AES-GCM, Reed-Solomon parity, and framing). NARCIS achieves 3–4 bpc nominal at K=8–16 and 0.184–1.213 bpc net. Prior systems reporting 10–15 bpc measure nominal capacity without authentication overhead; adding a comparable protocol stack would reduce their net throughput by the same constant cost. The paper reports both metrics explicitly and does not conflate them.

Author declarations.

- The manuscript is original, has not been published, and is not under consideration elsewhere.
- All four authors approved the final manuscript and author order. WYSAWIS (cited in the related work) shares authors with this submission and is explicitly identified as such in the paper; it is used as published qualitative context, not as an experimental baseline.
- No competing interests and no external funding.
- No human subjects, personal data, or animals are involved. Implementation, fixed seeds, experimental results, and manuscript

sources are available at github.com/EkodeckStephane/NARCIS.

We believe NARCIS will interest TMM readers working on multimedia steganography, learned image representations, and multimedia security evaluation. We welcome the opportunity to provide any additional information the Associate Editor may require.

Yours sincerely,

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